Practice workbook · Calculating fractions

Writing a group over its total and reducing it to lowest terms.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-14

1. Question 3 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

3 Davina investigated the relationship between the time spent on psychology homework and end of topic test scores.

Davina asked her students to write down the average time (in minutes) they spent on their homework each week and she also recorded their end of topic test scores (out of 30 marks).

Davina's results are shown in **Table 3**.

Participant	Average time spent on homework (in minutes)	End of topic test scores (out of 30 marks)
A	50	17
B	25	10
C	30	15
D	20	10
E	55	25
F	15	9
G	80	30
H	50	19
I	60	27
J	20	14

Table 3

(a) Calculate the range for the **end of topic test scores**.

(1)

Range .....

(b) Calculate how many students spent **more than** 30 minutes on their homework as a fraction of all students.

You must give your answer in the lowest form.

(2)

Fraction .....

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P 7 8 7 3 4 A 0 8 5 2

1. Question 3 · part b · 2 marks

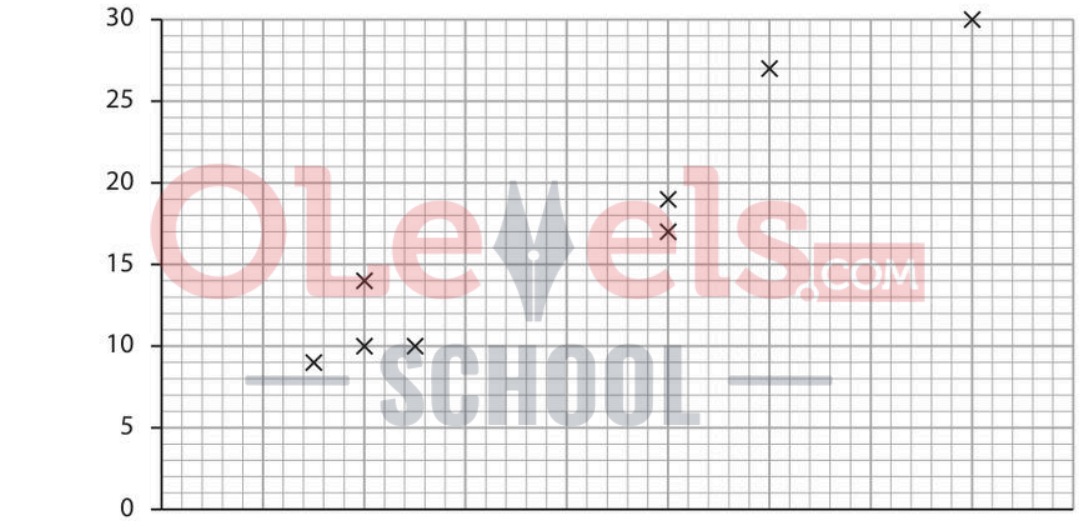
2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

(c) Complete the scatter diagram for participants C and E using the data in **Table 3**, including the label and scale for the x-axis.

(3)

A scatter diagram to show the relationship between average time spent on homework and end of topic test score



(Total for Question 3 = 6 marks)

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Working and answer

2. Question 2 · part b · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

2 Winston investigated gender differences in pro-social and anti-social behaviour. He placed a mobile phone on a seat near the local police station and recorded the responses of male and female passers-by to the mobile phone.

(a) Name the independent variable (IV) for Winston's investigation.

(1)

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.....

(b) The results for Winston's investigation are shown in **Table 2**.

	Number of male passers-by	Number of female passers-by
Took the mobile phone to the police station	3	7
Walked away with the mobile phone	8	8
Walked past the mobile phone	7	15

Table 2

Calculate how many participants were male as a fraction of all participants.

You must give your answer in the lowest form.

(2)

Males as a fraction of all participants .....

(c) Winston claims that females are more pro-social than males.

Give **one** way that the data in **Table 2** could support Winston's claim.

(1)

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(Total for Question 2 = 4 marks)

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2. Question 2 · part b · 2 marks · Working and answer

Working and answer

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3. Question 1 · part c · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part c only.

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1 Zak is investigating internet addiction. He designs a questionnaire that includes the following question:

How often do you use social media? Please tick one answer.

☐ Less than five times each day.
☐ Between five and ten times each day.
☐ More than ten times each day.

Figure 1

(a) Identify the type of question shown in **Figure 1**.

(1)

☒ **A** Unstructured question

☒ **B** Open-ended question

☒ **C** Closed-ended question

☒ **D** Semi-structured question

(b) State the type of data Zak will gather from his question in **Figure 1**.

(1)

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.....

(c) Zak found that nine out of fifteen participants used social media more than ten times each day.

Calculate the number of participants who used social media more than ten times each day as a fraction of all participants.

You must give your answer in the lowest form.

(1)

Fraction .....

(Total for Question 1 = 3 marks)

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3. Question 1 · part c · 1 mark · Working and answer

Working and answer

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4. Question 6 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

6 John researched the effect of a new drug treatment for depression. He had 60 patients with a diagnosis of depression that were split into two conditions.

- Condition A: 20 patients received a placebo sugar pill.
- Condition B: 40 patients received the new drug treatment.

In Condition A, 20% of patients reported a decrease in their symptoms of depression.
In Condition B, 85% of patients reported a decrease in their symptoms of depression.

(a) Calculate how many patients received the placebo sugar pill as a fraction of all patients.

You must give your answer in the lowest form.

(2)

Fraction .....

(b) Calculate the number of patients in **Condition A** who reported a decrease in their symptoms.

(1)

Number of patients .....

(c) John concludes that the new drug treatment is a more effective treatment for depression than a placebo sugar pill.

Give **one** way that John's findings could support this conclusion.

(1)

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(Total for Question 6 = 4 marks)

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4. Question 6 · part a · 2 marks · Working and answer

Working and answer

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5. Question 2 · part a · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a only.

2 Toby is investigating the impact of damage to the left hemisphere of the brain.

He asked patients with damage to their left hemisphere to complete a questionnaire about the impact it has had on them.

His results are shown in **Table 2**.

Patient	Language difficulty	Limited mobility in the right arm	Limited mobility in the right hand	Difficulties with the right visual field
A	Yes	No	No	Yes
B	Yes	Yes	No	Yes
C	Yes	Yes	No	No
D	Yes	Yes	Yes	Yes
E	Yes	Yes	Yes	No

Table 2

(a) Calculate how many patients had difficulties with the right visual field as a fraction of all patients.

(1)

Fraction .....

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5. Question 2 · part a · 1 mark

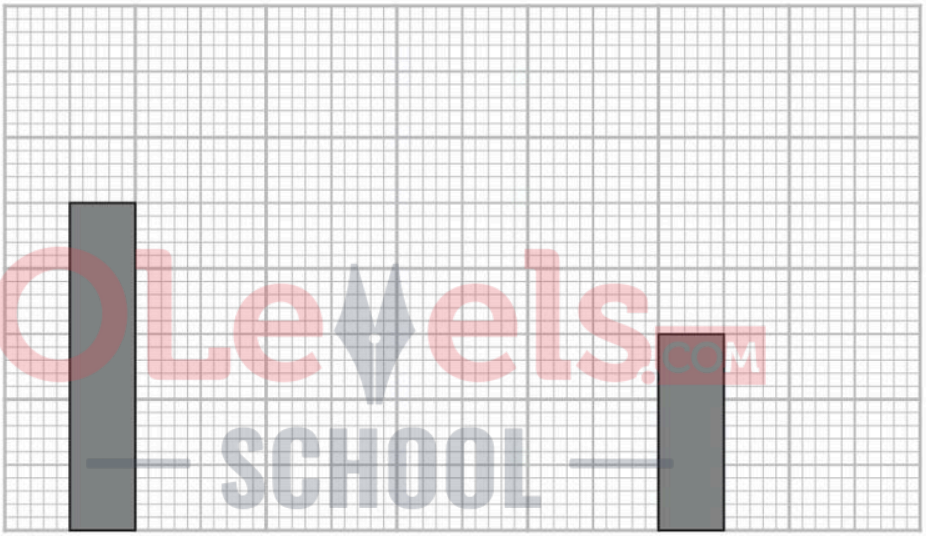
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Further practice · Attempt part a only.

(b) Complete the bar chart using the data in **Table 2**, including the label and scale for the y-axis.

(3)

A bar chart to show the number of patients reporting impacts from damage to the left hemisphere of the brain



Impact of damage to the left hemisphere

(Total for Question 2 = 4 marks)

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Working and answer

6. Question 3 · part a · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a only.

3 Quentin is investigating the capacity of short-term memory. He showed 20 objects to 10 participants for 30 seconds. Quentin then asked them to recall as many as they could.

His results are shown in **Table 2**.

Participant	Number of objects recalled
A	10
B	9
C	8
D	8
E	6
F	7
G	8
H	9
I	10
J	11

Table 2

(a) Calculate how many participants recalled nine objects as a fraction of all participants.

You must give your answer in the lowest form.

(1)

Fraction .....

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P 7 2 3 8 3 A 0 8 4 8

6. Question 3 · part a · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a only.

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(b) Calculate the median for the number of objects recalled.

(2)

Median .....

(c) Calculate the mode for the number of objects recalled.

(1)

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Mode .....

(d) State how Quentin would calculate the mean score for the number of objects recalled.

(1)

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(Total for Question 3 = 5 marks)

6. Question 3 · part a · 1 mark · Working and answer

Working and answer

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7. Question 4 · part b · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part b only.

4 Pedro was investigating whether schemas have an influence on memory accuracy. He decided to test four students from his psychology class.

He placed 10 typical objects, such as a chair, and 10 unusual objects, such as a laundry basket, in a classroom. The students had to enter the room and look around for one minute.

Pedro recorded the number of typical objects and unusual objects students could remember.

The results of Pedro's investigation are shown in **Table 1**.

	Number of typical objects recalled (out of 10)	Number of unusual objects recalled (out of 10)
Student A	8	3
Student B	10	5
Student C	9	5
Student D	5	2

Table 1

(a) Calculate the percentage of students who recalled **more than five typical objects**, out of all the students in the investigation.

(1)

Percentage .....

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P 7 5 5 2 4 A 0 6 4 8

7. Question 4 · part b · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part b only.

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(b) Calculate how many students recalled **less than five unusual objects** as a fraction of all students.

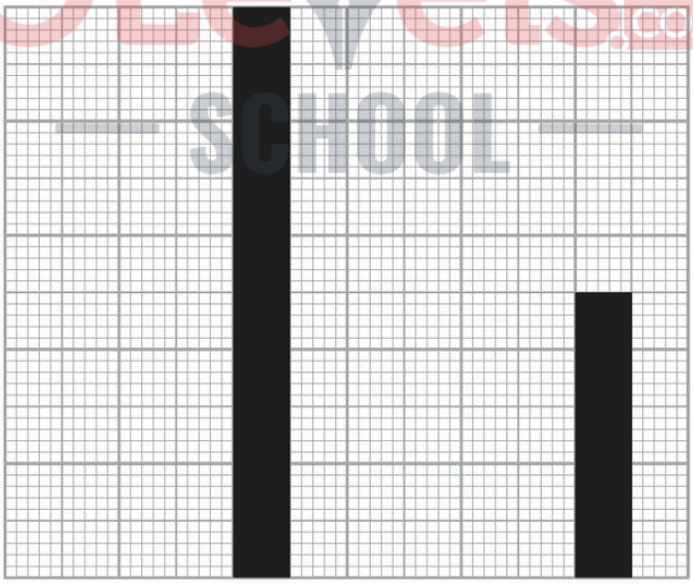
(1)

Fraction .....

(c) Complete the bar chart using the data in **Table 1**, including the label and scale for the y-axis.

(3)

A bar chart to show the number of typical objects recalled by the students



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P 7 5 5 2 4 A 0 7 4 8

7

Turn over

7. Question 4 · part b · 1 mark · Working and answer

Working and answer

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